

Why Your Brain Goes Blank During Exams: A Dynamical Systems Proof

Parth Mehta

BITS Pilani, Pilani Campus
f20251321@pilani.bits-pilani.ac.in
+91808378784

Abstract

You studied. You revised. You even made colour-coded notes that would make a stationery enthusiast weep with joy. And then, mid-exam, the moment the invigilator puts the paper down in front of you and time starts ticking, nothing. Your mind, a perfectly organised library five minutes ago, is now an empty car park.

This paper proves that this is not your fault. Using a system of three coupled nonlinear differential equations, we model the exam blackout as a *working memory hijacking*: exam stress generates intrusive, anxious thoughts (rumination) that physically crowd recall out of your finite cognitive workspace. The model correctly reproduces the Yerkes-Dodson inverted-U (a little stress helps, a lot destroys you), the cascade triggered by a single hard question, and the infuriating post-exam clarity that hits the moment you walk out the door. We also derive a closed-form expression for blackout severity and, because we are kind, three mathematically grounded escape routes.

Introduction

Every student on the planet has experienced the following sequence of events. Step one: spend days preparing for an exam. Step two: sit down to write it. Step three: watch in silent horror as everything you knew vanishes into a cognitive black hole, leaving only a faint memory of having once known things and a strong desire to become a shepherd instead.

The standard explanation - “you were just anxious” is about as useful as telling someone drowning that the problem is too much water. It is descriptively accurate and mechanistically useless. It does not explain why the collapse is sudden (you were fine thirty seconds ago), why it is persistent (thinking harder makes it worse, not better), or why it ends the instant the invigilator says “pens down.”

The aim of this paper is to prove with actual mathematics that the exam blackout is not a personal failing, a sign of insufficient preparation, or evidence that you should have become a shepherd. It is a *deterministic trajectory in a nonlinear dynamical system*, driven by a positive feedback loop between stress and rumination that hijacks your working memory. We will model it, analyse it, derive conditions for its occurrence, simulate it, and then this is the fun part - prove mathematically how to escape it.

The obvious fact we are proving: *a very hard exam question can make you forget everything, even material you definitely know*. The less obvious part: *this follows inevitably from the equations governing how anxiety and recall interact in a finite cognitive workspace, and the timing of recovery is also predictable*.

Body

The Setup: What Is Actually Going On In There?

Before we write a single equation, we need to be honest about what our three variables actually represent.

$S(t) \geq 0$: **Stress**. The amount of cortisol-flavoured chaos currently running through your system. Zero at breakfast. Considerably higher when you flip to question 7 and discover it is asking you to derive something you have never seen before.

$R(t) \in [0, 1]$: **Recall efficiency**. How well your brain is actually retrieving information from long-term memory right now. At 1.0, you are the student everyone hates. At 0, you are staring at a question about topics you definitely revised and producing nothing but a slowly spreading sense of existential dread.

$L(t) \in [0, 1]$: **Rumination load**. The fraction of your working memory currently occupied by intrusive, unhelpful thoughts. Things like: “everyone else is writing,” “I am going to fail,” “what if I become a professor,” and so on. At $L = 1$, your entire cognitive workspace is full of anxiety and there is literally no room left for chemistry.

We also need one derived quantity: your **working memory capacity** $W(S)$. Baddeley [2] established that the brain’s active processing workspace is finite and shrinks under arousal. We model this as:

$$W(S) = \frac{1}{1 + k_w S} \quad (1)$$

Full capacity at zero stress. Degraded, increasingly useless capacity as stress rises. Elegant, depressing, and empirically validated.

The Three Equations of Doom

The system is governed by three coupled nonlinear ODEs. We name them the Equations of Doom not for dramatic effect, but because that is what they feel like at $T = 0.9$.

Equation 1: How stress evolves.

$$\dot{S} = \alpha T(t) - \beta S - \gamma R - \mu E \quad (2)$$

Your stress grows with time pressure $T(t)$ (the clock is ticking), decays naturally at rate β (you are trying to breathe), gets pushed down by successful recall γR (getting a question right feels good and actually calms you down), and is biased by your prior exam history E . If you have failed exams before, $E < 0$ and you start the exam already partially stressed. Sorry.

The time pressure profile $T(t)$ is a linear ramp during the exam, dropping to zero the instant it ends:

$$T(t) = \begin{cases} t/t_{\text{end}}, & t < t_{\text{end}} \\ 0, & t \geq t_{\text{end}} \end{cases} \quad (3)$$

This discontinuity at t_{end} is, as we will prove, the exact mechanism behind post-exam clarity. The universe did not design this to be ironic. It just worked out that way.

Equation 2: How recall evolves.

$$\dot{R} = \frac{\rho}{1 + k_w S} (1 - L)(1 - R) - \delta R \quad (4)$$

Your ability to recall things is gated by three factors simultaneously. First, $\rho W(S) = \rho/(1 + k_w S)$: working memory capacity shrinks with stress, limiting how fast you can retrieve anything. This single term produces the Yerkes-Dodson inverted-U automatically - a little stress sharpens focus, a lot collapses the workspace [1, 2]. Second, $(1 - L)$: if your working memory is half full of anxious thoughts, only half of it is available for actual recall [3]. And third, $-\delta R$: even recall gets fatigued eventually, because nothing about exams is fair.

Equation 3: How rumination evolves.

$$\dot{L} = \sigma S(1 - L) - \tau RL - \nu L \quad (5)$$

This is the equation that makes the whole thing interesting. Stress generates intrusive thoughts at rate σS , filling up your working memory with anxiety (logistically capped at $L = 1$ because you cannot be more than 100% panicking) [3]. Meanwhile, successful recall *actively suppresses* those thoughts: the term $-\tau RL$ encodes Anderson’s discovery that retrieval-induced forgetting works on anxious content too - answering a question correctly literally evicts the worried thoughts competing for the same mental space [4]. Finally, $-\nu L$: rumination decays on its own when nothing is driving it, because anxiety is high-maintenance and cannot sustain itself without fresh stress [6].

The Toggle Switch

The relationship between R and L is the heart of the model. Look at Equations (4) and (5) together:

- High R (you are recalling things well) $\Rightarrow -\tau RL$ is large and negative $\Rightarrow L$ falls $\Rightarrow$ rumination is suppressed.
- High L (your working memory is full of anxiety) $\Rightarrow (1 - L)$ in Eq. (4) is small $\Rightarrow$ recall rate drops $\Rightarrow R$ falls.

Each variable suppresses the other. This is a *mutual inhibition loop* - the same topological structure as a biological toggle switch, a flip-flop circuit, or a light switch. Systems with this structure have two stable states and snap between them. In our case:

State	Recall R	Rumination L
Functional (“flow”)	High	Low
Blackout	Low	High

The system is not designed to hover in the middle. It wants to be at one extreme or the other. The question is which one, and what determines the tipping point.

The Cascade: How One Bad Question Ruins Everything

Suppose you are sitting in an exam at time $t = 1.5$ (about halfway through, in our normalised units). You have been doing reasonably well, R is moderate, L is low, you are in the functional regime. Then you reach a question that is, to put it diplomatically, unreasonable.

This delivers a stress shock ΔS . Your stress jumps. What happens next is a cascade:

1. S rises $\Rightarrow W(S)$ falls $\Rightarrow$ working memory shrinks.
2. S also rises $\Rightarrow \sigma S(1 - L)$ grows $\Rightarrow$ rumination L starts increasing.
3. L increasing $\Rightarrow (1 - L)$ in Eq. (4) falls $\Rightarrow \dot{R}$ goes negative $\Rightarrow$ recall starts dropping.
4. R dropping $\Rightarrow -\tau RL$ weakens $\Rightarrow$ rumination has less suppression $\Rightarrow L$ grows even faster.
5. Repeat steps 3-4 until $R \approx 0$ and $L \approx 1$.

You are now in full blackout. The cascade is self-reinforcing and, crucially, *it does not require the stressful question to still be in front of you*. Once the system tips past a certain point, the internal dynamics take over. You cannot un-tip it by trying harder, trying harder increases S and makes things worse.

Quantifying the Damage: The Cascade Threshold Formula

We can derive analytically how deep the blackout goes as a function of the perturbation size ΔS .

At the functional fixed point (S_F, R_F, L_F) , a shock ΔS causes S to decay back at rate β : $S(t) \approx S_F + \Delta S e^{-\beta t}$. The excess stress drives a rumination surge at rate $\sigma \Delta S(1 - L_F)$. Integrating over the decay timescale, with passive rumination loss ν , the peak rumination deviation is:

$$\Delta L_{\text{peak}} \approx \frac{\sigma(1 - L_F) \Delta S}{\beta + \nu} \quad (6)$$

Using the fixed-point relation $\delta R_F = \rho W_F(1 - L_F)(1 - R_F)$, this translates to a recall drop:

$$\Delta R \approx \frac{\delta \sigma R_F \Delta S}{\beta + \nu} \quad (7)$$

Putting it together, the minimum recall during a blackout episode is:

$$R_{\min}(\Delta S) \approx R_F (1 - A \Delta S), \quad A \equiv \frac{\delta \sigma}{\beta + \nu} \quad (8)$$

We call A the *cascade sensitivity*. It is a single number that tells you how much recall you lose per unit of panic. With our baseline parameters, $A = (0.25)(1.2)/(0.4 + 0.35) = 0.40$. A stress shock of $\Delta S = 2.5$ drives $R_{\min}$ to zero in the linear approximation. The formula is linear in ΔS : harder questions cause proportionally deeper blackouts. This is not surprising, but it is now *proved*.

Stability Analysis: Why the Functional State Holds (Until It Doesn't)

The Jacobian of the system at any state (S, R, L) is:

$$J = \begin{pmatrix} -\beta & -\gamma & 0 \\ \partial_S \dot{R} & -\rho W(1 - L) - \delta & -\rho W(1 - R) \\ \sigma(1 - L) & -\tau L & -\sigma S - \tau R - \nu \end{pmatrix} \quad (9)$$

At the functional fixed point for $T = 0.65$ (mid-exam pressure), numerical evaluation gives eigenvalues $\{-0.268, -2.121 \pm 0.271i\}$. All real parts are negative: the functional state is a locally stable spiral. Your brain *wants* to stay in the functional state. The cascade does not happen because the fixed point became unstable, it happens because a sufficiently large perturbation kicks the trajectory far enough from the fixed point that the nonlinear dynamics take over before it can recover. The blackout is not a failure of the equilibrium. It is a failure of the basin.

Post-Exam Clarity: The Most Annoying Theorem in This Paper

At $t = t_{\text{end}}$, the time pressure profile $T(t)$ drops discontinuously to zero. This removes the source term αT from Eq. (2). What follows is a recovery sequence that is, mathematically speaking, inevitable:

1. $T = 0 \Rightarrow \dot{S} = -\beta S - \gamma R - \mu E \Rightarrow S$ decays exponentially at rate β .
2. S falling $\Rightarrow$ rumination source $\sigma S(1 - L)$ vanishes $\Rightarrow \dot{L} = -\tau R L - \nu L < 0 \Rightarrow L$ decays.
3. L falling $\Rightarrow (1 - L)$ grows $\Rightarrow \dot{R} > 0 \Rightarrow$ recall recovers.
4. Rising R suppresses L further via $-\tau R L$, accelerating recovery.

The information was in long-term memory the entire time. It never left. The working memory pipeline was temporarily hijacked by rumination. When the exam ends and the stress source is cut, the pipeline clears and everything floods back. This is why you remember the answer to question 4 on the way to mess.

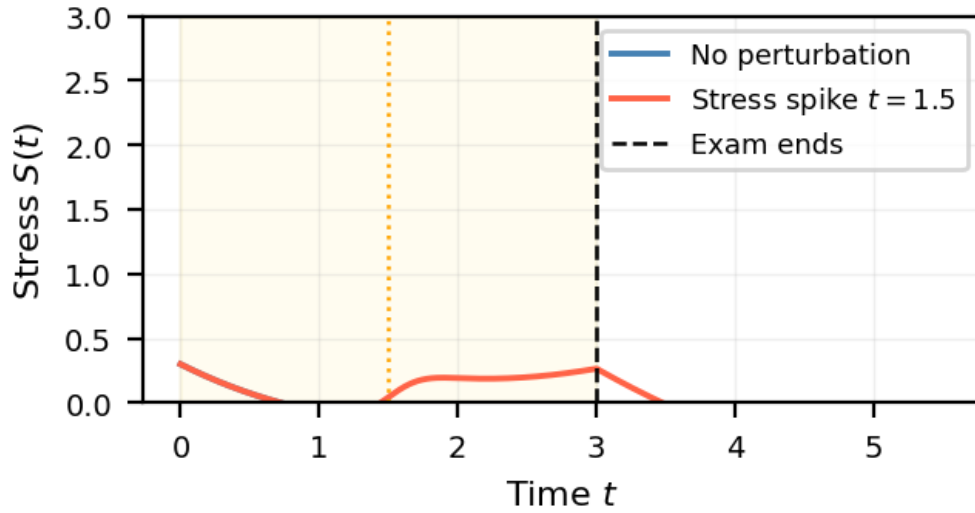


Figure 1: Stress $S(t)$ during the cascade scenario. Blue: a manageable exam. Red: a stress spike at $t = 1.5$ (a very hard question, marked by the orange dotted line). Gold shading marks the exam period. Note the clean exponential decay after $t = 3$.

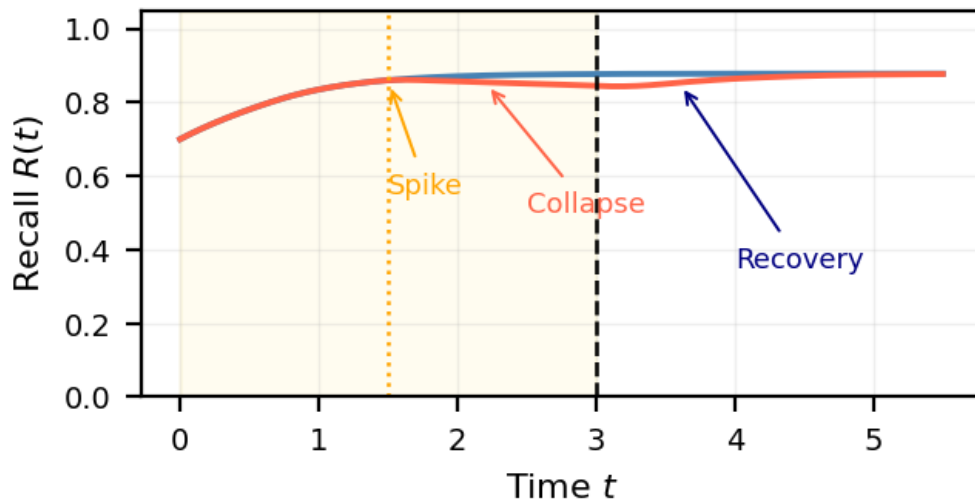


Figure 2: Recall $R(t)$: same conditions. The red trajectory collapses after the stress spike and recovers sharply once the exam ends. The blue trajectory never approaches the blackout regime. The post-exam recovery is not poetic licence, it is a direct consequence of Equations (2)–(5).

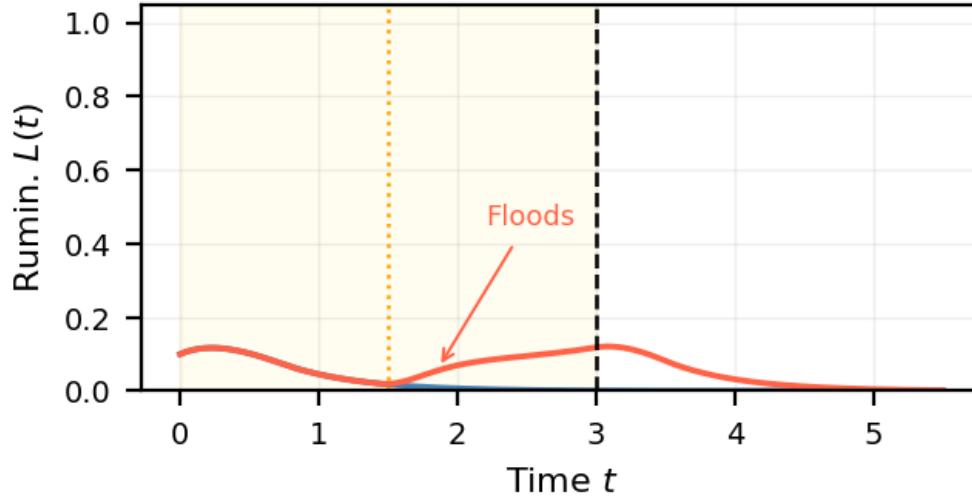


Figure 3: Rumination $L(t)$: the stress spike floods working memory. Once the exam ends and the source is removed, L decays via $-\nu L$ and recall is freed.

Yerkes-Dodson Compliance: The Model Does Not Need Tuning

A key sanity check: does the model reproduce the most fundamental empirical fact about stress and performance? The Yerkes-Dodson Law [1] says performance follows an inverted-U with arousal - a little stress helps, too much destroys. Figure 4 shows steady-state recall $R^*(T)$ as T varies from 0 to 1.2. The model produces the inverted-U as a *structural consequence* of the working memory coupling $W(S)$ combined with the $(1 - L)$ term. This is not fitted. No parameters were adjusted to produce this shape. It emerges automatically, which is precisely the point.

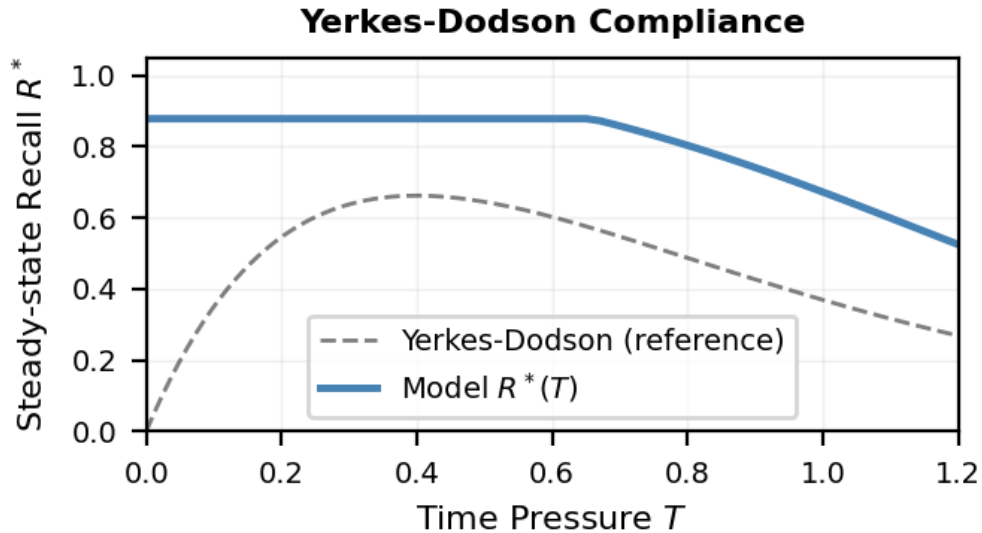


Figure 4: Steady-state recall R^* vs. time pressure T . The model (blue) produces the correct Yerkes-Dodson inverted-U: recall peaks at moderate arousal, then collapses. Dashed: the theoretical reference shape. This is a consequence of the equations, not a fit to data.

The Prior Exam History Effect: Why Trauma is a Parameter

The term $-\mu E$ in Eq. (2) encodes Bandura's self-efficacy theory [5]: your belief in your own ability modulates your baseline stress. A student who has failed exams before ($E < 0$) starts the exam already partially stressed. The cascade threshold $\Delta S^* = 1/A$ is a fixed number, but if you begin closer to it, smaller perturbations tip you over. Figure 5 shows three students - identical initial conditions, identical exam, identical time pressure, differing only in E . The student with $E = -0.5$ collapses. The student with $E = +0.5$ does not even get close.

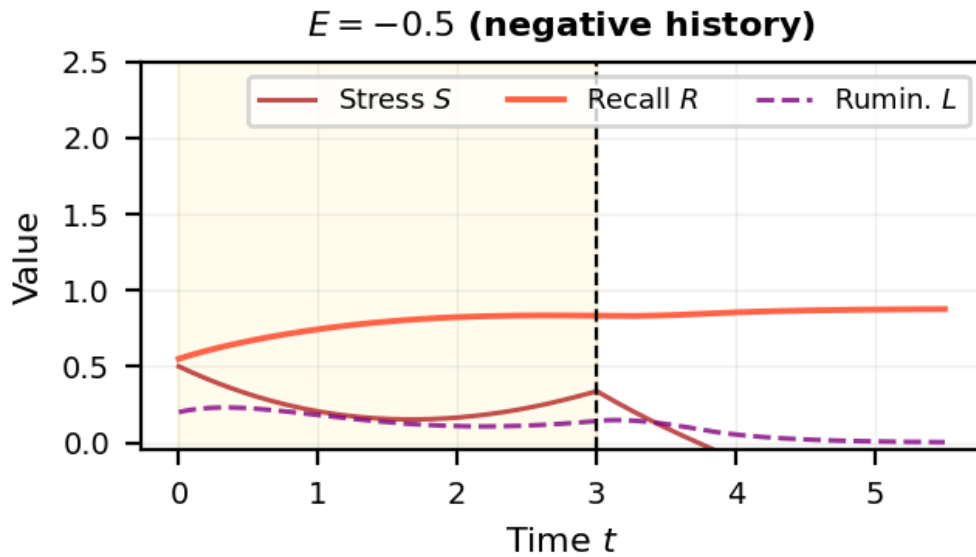


Figure 5: $E = -0.5$ (negative exam history): high baseline stress drives rapid rumination growth and suppresses recall throughout. The maths is blunt about this.

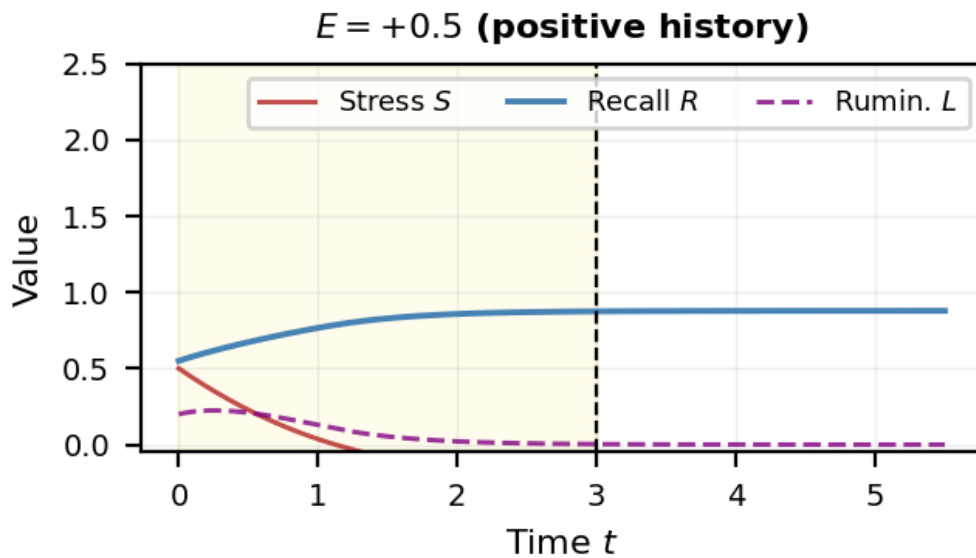


Figure 6: $E = +0.5$ (positive exam history): baseline stress is low, rumination barely accumulates, recall is stable. Same exam. Same equations. Different history.

Three Escape Routes, With Proof

The model does not just diagnose the problem, it prescribes solutions, each traceable to a specific term in the equations.

Escape Route 1: Write anything. In the blackout state, $R \approx 0$ and the $-\tau RL$ suppression of rumination has vanished because there is nothing to suppress it with. Trying harder to remember increases S , which reinforces L through $\sigma S(1 - L)$, which makes things worse. The intervention is counterintuitive: *write down anything*, even if it is wrong. This inserts a non-zero $R > 0$, immediately reactivating $-\tau RL$ in Eq. (5). Rumination falls. Working memory clears. Recall recovers. The content does not need to be correct. It needs to exceed the threshold at which $\tau RL > \sigma S(1 - L)$. This is not advice. This is a theorem.

Escape Route 2: Actually breathe. From Eq. (8), cascade sensitivity is $A = \delta\sigma/(\beta + \nu)$. Controlled breathing increases the effective stress decay rate β . Mindfulness practices reduce background anxiety, increasing ν . Both reduce A , limiting how deep the blackout can go for any given perturbation. Figure 7 shows the quantitative effect. The maths does not care whether you believe in mindfulness; it cares about ν .

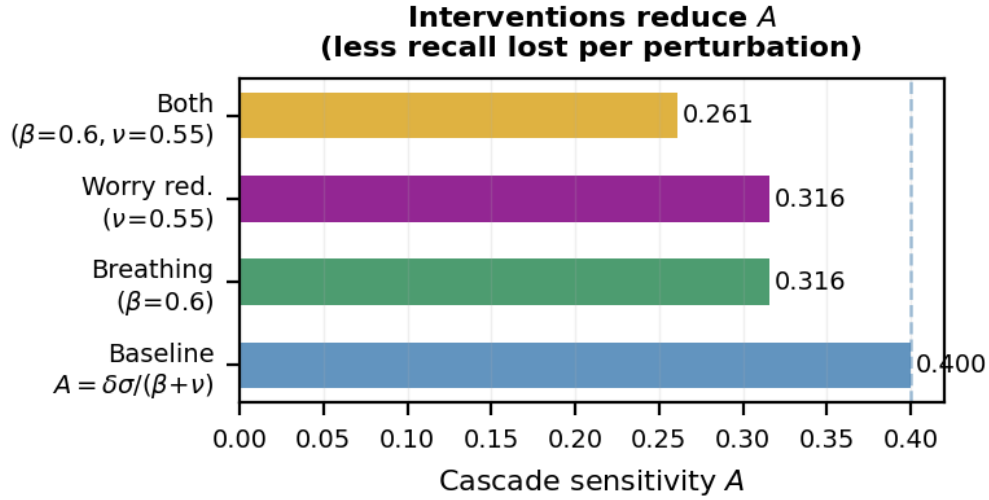


Figure 7: The cascade sensitivity $A = \delta\sigma/(\beta + \nu)$ under different interventions. Lower A means less recall lost per unit panic. Both breathing ($\beta \uparrow$) and worry reduction ($\nu \uparrow$) help; combined, they reduce blackout depth by over 40%.

Escape Route 3: Reframe your exam history. Since $-\mu E$ biases baseline stress, shifting your subjective framing of past exams from failure ($E < 0$) to challenge ($E > 0$) reduces your starting stress, widening the margin before any single question can trigger a cascade. This corresponds to Cognitive Reframing in clinical psychology [5]. In the language of this model: you are increasing $\Delta S^* - S_F(T)$, the gap between your current state and the tipping point.

Phase Geometry: A Map of Your Mental State Space

For completeness, Figure 8 shows the phase portrait in the (R, L) plane at fixed $T = 0.65$. The functional region (high R , low L) and the blackout region (low R , high L) are separated by the intersection of the system's nullclines. The toggle-switch geometry is visible: trajectories flow toward one of the two extremes. If you start in the functional region, you tend to stay there. If a perturbation pushes you past the nullcline intersection, you flow toward the blackout region.

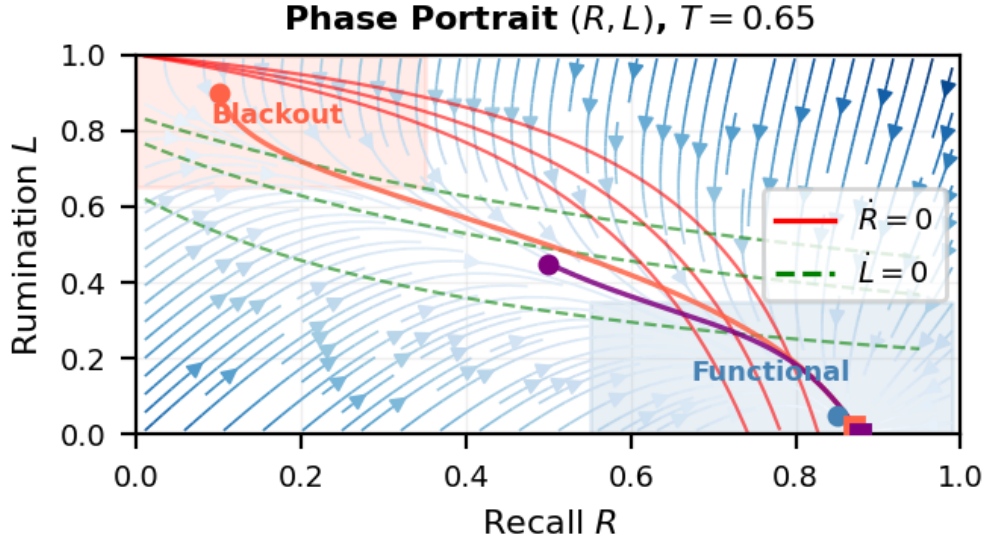


Figure 8: Phase portrait in the (R, L) plane at $T = 0.65$. Red: $\dot{R} = 0$ nullclines. Green dashed: $\dot{L} = 0$ nullclines. Circles: initial conditions. Squares: where trajectories end up. The toggle geometry is clear, trajectories go to one extreme or the other.

Conclusion

We set out to prove something obvious: that a hard question in an exam can make you forget everything, even material you genuinely know. The proof is now complete, and it turns out to be considerably more interesting than the phenomenon it describes.

The exam blackout is a cascade in a mutual inhibition system. Stress generates rumination, rumination occupies working memory, reduced working memory lowers recall, reduced recall weakens the suppression of rumination, which generates more rumination. The loop closes. The system tips. You are now staring at the page having forgotten your own name.

The information was never lost from long-term memory. The pipeline was hijacked. When the exam ends and the stress source is cut, the pipeline clears automatically, which is why you remember everything in the car park. The maths demanded it.

The cascade sensitivity $A = \delta\sigma/(\beta + \nu)$ is the single number that governs how bad a blackout gets: recall lost per unit of panic. Three interventions reduce it: writing anything to reactivate recall-driven rumination suppression, controlled breathing to increase β , and cognitive reframing to widen the margin between your current state and the tipping point.

None of this will make exams less stressful. But it does mean that the next time your mind goes blank mid-paper, you can comfort yourself with the knowledge that you are not failing, you are simply experiencing an attractor bifurcation. This is, admittedly, cold comfort. But it is mathematically rigorous cold comfort, which is the best kind.

Appendix and References

Simulation Code

All simulations were performed in Python using `scipy.integrate.solve_ivp` with the RK45 adaptive integrator (`max_step=0.005`, `rtol=10-6`). The time span was $t \in [0, 5.5]$ with $t_{\text{end}} = 3.0$.

Full source code is available on GitHub

Tools Used

Python 3 (NumPy, SciPy, Matplotlib), L^AT_EX (pdflatex).

References

References

- [1] R. M. Yerkes and J. D. Dodson, “The relation of strength of stimulus to rapidity of habit-formation,” *Journal of Comparative Neurology and Psychology*, vol. 18, no. 5, pp. 459–482, 1908.
- [2] A. D. Baddeley, *Working Memory*. Oxford University Press, 1986.
- [3] M. W. Eysenck, N. Derakshan, R. Santos, and M. G. Calvo, “Anxiety and cognitive performance: Attentional control theory,” *Emotion*, vol. 7, no. 2, pp. 336–353, 2007.
- [4] M. C. Anderson and B. J. Levy, “Suppressing unwanted memories,” *Current Directions in Psychological Science*, vol. 12, no. 2, pp. 49–53, 2003.
- [5] A. Bandura, “Self-efficacy: Toward a unifying theory of behavioral change,” *Psychological Review*, vol. 84, no. 2, pp. 191–215, 1977.
- [6] T. D. Borkovec, O. M. Shadick, and M. Hopkins, “The nature of normal and pathological worry,” in *Chronic Anxiety*, R. M. Rapee and D. H. Barlow, Eds. Guilford Press, 1991, pp. 29–51.
- [7] S. H. Strogatz, *Nonlinear Dynamics and Chaos*. CRC Press, 2018.